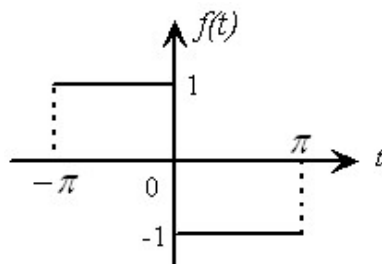


6-1 如图所示矩形波，请用下列函数近似

$$f(t) = C_1 \sin t + C_2 \sin 2t + \dots + C_n \sin nt$$



任一函数在给定的区间内可以用在此区间的完备正交函数集表示，但若只取函数集中的有限项，或者正交函数集不完备，则只能得到近似的表达式。

$$\begin{aligned} C_n &= \frac{\int_{-\pi}^{\pi} f(t) \sin ntdt}{\int_{-\pi}^{\pi} \sin^2 ntdt} \\ &= \frac{-\int_0^{\pi} \sin ntdt}{\int_0^{\pi} \sin^2 ntdt} \\ &= \frac{\frac{1}{n} \cos nt \Big|_0^{\pi}}{\frac{1}{2} \left[t \Big|_0^{\pi} - \frac{1}{2n} \cos 2nt \Big|_0^{\pi} \right]} = \frac{2}{n\pi} [(-1)^n - 1] \end{aligned}$$

$$C_1 = -\frac{4}{\pi}, C_2 = 0, C_3 = -\frac{4}{3\pi}, C_4 = 0, \dots$$

6-2 测 P115 的 2-11

$$\omega = 15, f = \frac{15}{2\pi};$$

$$y(t) = 100 + 109.77 \cos(15t - 10.46)$$

6-3 测 P115 的 2-12

$$(1) a_0 = a_n = 0;$$

$$b_n = \frac{(-1)^{n+1} * 2}{n\pi};$$

$$(2) a_0 = a_n = 0;$$

$$b_n = \frac{[1 - (-1)^n] * 2A}{n\pi}$$

6-4

已知 $f(t) \Leftrightarrow F(j\omega)$ ，求下列信号的傅里叶变换。

(1) $tf(2t)$

(2) $(t-2)f(t)$

(3) $(t-2)f(-2t)$

(4) $t \frac{df(t)}{dt}$

(5) $f(1-t)$

(6) $(1-t)f(1-t)$

(7) $f(2t-5)$

(8) $tU(t)$

解：(1)

$$\because F[f(2t)] = \int_{-\infty}^{\infty} f(2t)e^{-j\omega t} dt$$

$$\therefore \frac{dF[f(2t)]}{d\omega} = \int_{-\infty}^{\infty} -tjf(2t)e^{-j\omega t} dt = F[-jtf(2t)]$$

$$\therefore \frac{dF[f(2t)]}{d\omega} \Leftrightarrow -jtf(2t) \quad j \frac{dF[f(2t)]}{d\omega} \Leftrightarrow tf(2t)$$

$$\therefore F[f(2t)] = \frac{1}{2} F(j\frac{\omega}{2})$$

$$\therefore tf(2t) \Leftrightarrow \frac{1}{2} jF'(j\frac{\omega}{2})$$

$$(2) (t-2)f(t) = tf(t) - 2f(t) \Leftrightarrow jF'(j\omega) - 2F(j\omega)$$

$$(3) (t-2)f(-2t) = tf(-2t) - 2f(-2t) \Leftrightarrow j\frac{1}{2} \frac{d}{d\omega} \left[F(-j\frac{\omega}{2}) \right] - 2 \times \frac{1}{2} F(-j\frac{\omega}{2})$$

$$(t-2)f(-2t) \Leftrightarrow j\frac{1}{2} F'(-j\frac{\omega}{2}) - F(-j\frac{\omega}{2})$$

(4) 因有

$$-jtf(t) \Leftrightarrow \frac{dF(j\omega)}{d\omega}$$

两边同时求导

$$-jt \frac{df(t)}{dt} - jf(t) = \frac{d\left(\frac{dF(j\omega)}{d\omega}\right)}{d\omega} = \omega j \frac{dF(j\omega)}{d\omega}$$

$$-t \frac{df(t)}{dt} - f(t) = \omega \frac{dF(j\omega)}{d\omega}$$

$$-t \frac{df(t)}{dt} \Leftrightarrow \omega \frac{dF(j\omega)}{d\omega} + F(j\omega)$$

所以有：

$$t \frac{df(t)}{dt} \Leftrightarrow -[\omega F'(\omega) + F(\omega)]$$

(5)

$$\begin{aligned} F[f(1-t)] &= \int_{-\infty}^{\infty} f(1-t) e^{-j\omega t} dt = \int_{u=1-t}^{\infty} f(u) e^{-j\omega(1-u)} du \\ &= e^{-j\omega} \int_{-\infty}^{\infty} f(u) e^{j\omega u} du = e^{-j\omega} F(-j\omega) \end{aligned}$$

$$(6) \quad (1-t)f(1-t) = f(1-t) - tf(1-t)$$

$$\Leftrightarrow F(-j\omega) e^{-j\omega} - \left\{ j \frac{d}{d\omega} [F(-j\omega) e^{-j\omega}] \right\} = -jF'(-j\omega) e^{-j\omega}$$

$$\begin{aligned} F[f(1-t)] &= \int_{-\infty}^{\infty} f(1-t) e^{-j\omega t} dt = \int_{u=1-t}^{\infty} f(u) e^{-j\omega(1-u)} du \\ &= e^{-j\omega} \int_{-\infty}^{\infty} f(u) e^{j\omega u} du = e^{-j\omega} F(-j\omega) \end{aligned}$$

$$\frac{d(F[f(1-t)])}{d\omega} = \frac{\int_{-\infty}^{\infty} f(1-t) e^{-j\omega t} dt}{d\omega} = \int_{-\infty}^{\infty} -jtf(1-t) e^{-j\omega t} dt = F[-jtf(1-t)]$$

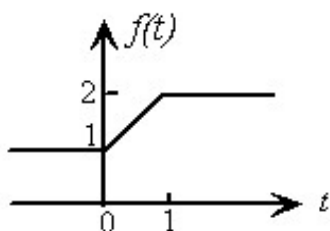
$$\therefore tf(1-t) \Leftrightarrow j \frac{d(F[f(1-t)])}{d\omega} = j \frac{d(e^{-j\omega} F(-j\omega))}{d\omega} = e^{-j\omega} F(-j\omega) + je^{-j\omega} F'(-j\omega)$$

$$\begin{aligned} \therefore F[(1-t)f(1-t)] &= F[f(1-t) - tf(1-t)] \\ &= F[f(1-t)] - F[tf(1-t)] = -je^{-j\omega} F'(-j\omega) \end{aligned}$$

$$(7) \quad f(2t-5) = f\left[2\left(t-\frac{5}{2}\right)\right] \Leftrightarrow \frac{1}{2} F\left(j\frac{\omega}{2}\right) e^{-j\frac{5}{2}\omega}$$

$$(8) \quad tU(t) \Leftrightarrow j \frac{d}{d\omega} \left[\pi\delta(\omega) + \frac{1}{j\omega} \right] = \pi j \delta'(\omega) - \frac{1}{\omega^2}$$

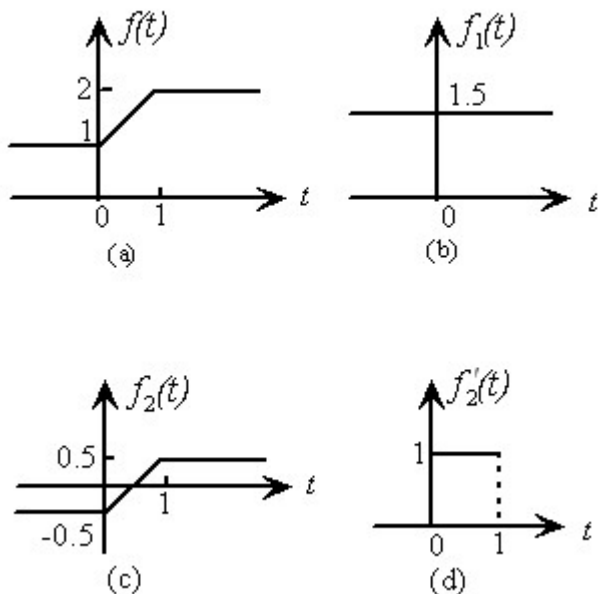
6-5 求如图函数的傅里叶变换。



解:

将 $f(t)$ 分解为 $f_1(t)$ 与 $f_2(t)$ 的叠加。即 $f(t) = f_1(t) + f_2(t)$

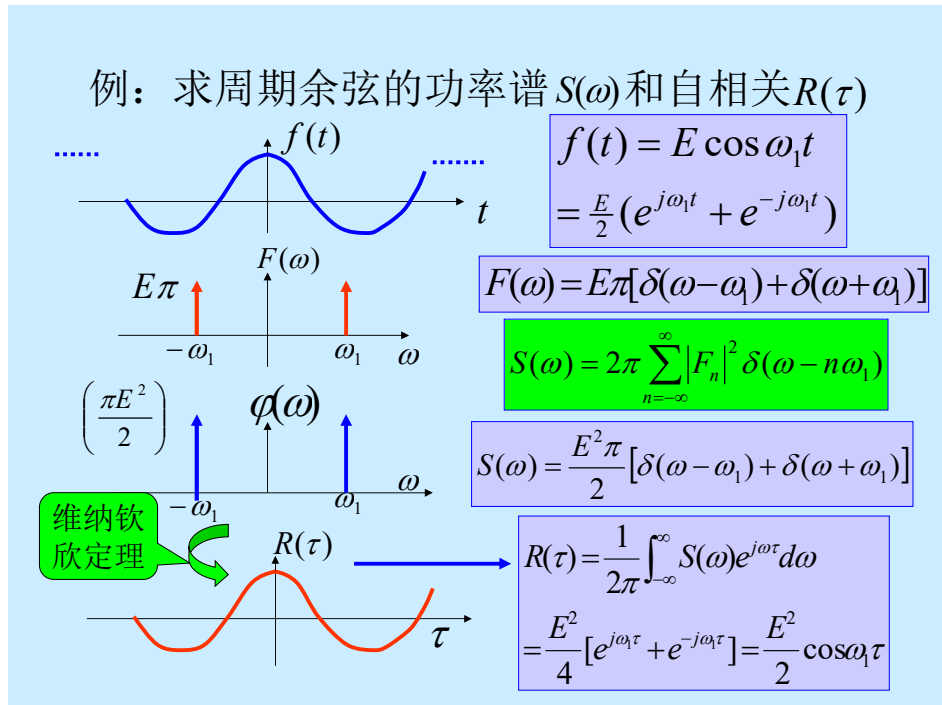
如图,



$$F(j\omega) = F_1(j\omega) + F_2(j\omega) = 3\pi\delta(\omega) + \frac{1}{j\omega} \text{Sa}\left(\frac{\omega}{2}\right) e^{-j\frac{1}{2}\omega}$$

也可以直接利用 $t*U(t)$ 和 $U(t)$ 的 Fourier 变换，不过结果要注意化简：

6-6 求周期余弦的功率谱 $f(t) = E \cos(\omega_1 t)$ 功率谱和自相关



6-7

$$F(\omega) = E^2 \tau^2 \text{Sa}^2\left(\frac{\omega\tau}{2}\right)$$